

Vinegar's Effect on Bacterial Growth in Milk: Lab Summary

Introduction

Our group (Anav, Evan, Ben) tested whether adding vinegar slows bacterial growth in milk. Methylene blue was the indicator: it turns white as bacteria use up the oxygen in the tube.

- **Purpose:** see the impact of vinegar on the growth of bacteria in milk.
- Bacteria grow naturally in untreated milk over time.
- Milk is already slightly acidic (pH about 6.5).
- Vinegar is much more acidic (pH about 2.5), so adding it lowers the pH of the milk.
- A low pH (an acidic solution) harms cells and can kill them.
- Methylene blue is blue when oxygen is present and turns white as oxygen is used up, so a faster colour change means faster bacterial growth.

Hypothesis

If we add vinegar to milk in a closed test tube, then bacterial growth will slow, and more vinegar will slow it further. This is because vinegar lowers the pH of the milk, and a low pH harms and kills bacterial cells.

Materials

- Vinegar (5%) and milk
- Methylene blue indicator (with dropper)
- Graduated cylinder x 1
- Test tubes x 3, test tube rack x 1
- 500 mL beakers and a water heater (water bath)
- Thermometer
- Timing device (a mobile phone)
- Camera and a plain background (whiteboard) for daily photos
- Safety glasses, closed-toe shoes, appropriate lab attire

Method

We added 5, 10 and 15 mL of vinegar to three tubes of milk, warmed them in a water bath, stored them in the dark, and photographed them daily until the indicator changed colour.

Variables

- **Independent variable:** volume of vinegar added (5, 10 and 15 mL), measured in a graduated cylinder from the bottom of the meniscus, always by the same person.
- **Dependent variable:** days for the tube to turn white, measured with daily photos at about the same time each day against a whiteboard background, then compared after the experiment finished.
- **Controls (comparison tubes):** a milk-only tube (15 mL milk, no vinegar) and a water-only tube with indicator (no milk). Both were class controls shared across groups.

Controlled variables

Controlled variable	Value	How it was kept constant
Type of milk	2% (to confirm)	All milk from the same bag
Volume of milk	15 mL	Graduated cylinder, same person poured, read at the bottom of the meniscus
Time in heated water	10 min	One timer; all 3 tubes in and out together; removal started at 9:58 so none came out early
Water bath temperature	___ °C (fill in)	Same water bath for all tubes
Vinegar concentration	5%	Same container for every sample
Methylene blue	10 drops	Same dropper, same person
Storage	Dark environment	All tubes kept together in the dark, same location
Tube cover	Rubber stopper (closed tube)	Same type on all 3 vinegar tubes
Photo conditions	Same background, about the same time daily	Whiteboard background

Step-by-step procedure

1. Put on safety glasses, closed-toe shoes and lab attire. Set up the test tube rack, graduated cylinder, 5% vinegar, milk, methylene blue dropper, thermometer and a timer (a phone).
2. Label three test tubes with tape and the initials of the group, near the top so the label does not fall off in the water. Mark them 5 mL, 10 mL and 15 mL of vinegar.
3. Using the graduated cylinder, measure 15 mL of milk for each tube. Read from the bottom of the meniscus, and have the same person pour every time. Milk for all tubes comes from the same bag.
4. Rinse the graduated cylinder, then measure 5 mL, 10 mL and 15 mL of 5% vinegar from the same container, one volume for each tube. The same person adds the vinegar to all three tubes.
5. Add 10 drops of methylene blue to each tube with the same dropper, always by the same person.
6. Close each tube with a rubber stopper (no holes) so it fits securely without sinking into the tube. Gently mix until the contents look even.
7. Prepare the water bath: heat water and pour it into 500 mL beakers. Check the temperature with the thermometer (___ °C). Use the thermometer only to measure, never to stir.
8. Place all three tubes into the water bath at the same time and start a 10 minute timer. Keep the tubes together for the whole time.
9. Start taking the tubes out at 9:58 so that they all stay in for at least the full 10 minutes. Remove them together.
10. Take a Day 1 photo of the three tubes against the whiteboard background, with the same lighting, and record all observations (colour, curdling, layers).
11. Store the tubes together in a dark environment, in a place all group members know.
12. Each day at about the same time, take the tubes out, photograph them against the same background, and record the colour of each tube and any curdling or layers. Put them back in the dark right away.
13. Continue until the control tube has turned white and the vinegar tubes have been checked on each day (5 days).
14. Compare the photos afterward to judge when each tube turned white.
15. Put all waste in the waste beaker and nothing down the sink. Wash hands and clean the workspace.

Comparison tubes (class controls)

- **Milk only:** 15 mL of milk and 10 drops of methylene blue, no vinegar. This shows how fast bacteria turn the indicator white without vinegar.
- **Water only:** water and methylene blue, no milk. This shows the indicator does not turn white without bacteria.
- Both were set up and photographed on the same days as our tubes and covered with foil.

Results

The milk-only control turned white by Day 3 to 4, while none of the three vinegar tubes turned white by the end of the experiment.

Vinegar added (mL)	Days to turn white
0 (milk-only control)	About 3 (mostly white Day 3, fully white Day 4)
5	Did not turn white (more than ___ days)
10	Did not turn white (more than ___ days)
15	Did not turn white (more than ___ days)
Water only (control)	Stayed blue all 5 days

Because the three vinegar tubes gave the same result, a line graph would show no trend, so a table and photos are used instead. Tube order in the photos is assumed to be 5, 10, 15 mL left to right (to confirm).

Control tubes (water only = D, milk only = E)

- Water only: dark navy blue on all 5 days, so no bacteria and no oxygen use.
- Milk only: bright blue on Day 1, lighter blue on Day 2, mostly white with a teal layer on Day 3, white with a thin blue band on Day 4, fully white with a small blue ring at the surface on Day 5.

Vinegar tubes: qualitative observations

- All three stayed blue at every daily check.
- The milk curdled in every tube: a lighter blue, clumpy layer separated from a darker blue, smoother liquid underneath.

- A dark blue layer stayed separate from the rest each day.
- The curdled layer looked larger in the tubes with more vinegar.
- Small dark specks collected near the stopper.

Insert the control photos (Day 1 to Day 5) and the vinegar tube photos here.

Discussion

1. Hypothesis: supported

The milk-only control turned white in about 3 days, but all three vinegar tubes stayed blue, so vinegar slowed bacterial growth. Caveat: none of the vinegar tubes changed, so we cannot say that more vinegar slowed growth more than less. We can only say that vinegar slowed it.

2. Sources of error

Systemic (built into the procedure or equipment)

- Daily photos limit the timing to about 1 day, so smaller differences cannot be detected.
- Graduated cylinder precision is limited (about 0.5 mL), which affects every vinegar volume.
- Adding vinegar dilutes the milk, so the 5, 10 and 15 mL tubes held different total volumes and different milk concentrations.
- Vinegar lowers the pH, which may change methylene blue's colour directly, so blue may not mean "no bacteria".
- Curdling made the liquid uneven, so the indicator could only be judged from the part of the tube we could see.
- pH was never measured, so we assumed the vinegar changed it.
- The control tubes were covered with foil, but the vinegar tubes had rubber stoppers, so oxygen exposure was not identical.
- Only one trial per condition.

Random (cannot be controlled and may vary between tubes)

- Judging "whiteness" and "blueness" by eye from photos is subjective.
- Lighting and camera angle varied slightly from day to day.
- Starting bacteria levels in each 15 mL sample may differ a little.

- Curds and liquid layers formed differently in each tube.
- Water bath temperature drifted slightly, and drop and pour sizes varied a bit.

Blunder (correctable errors; list only what actually happened)

- Photos taken at inconsistent times of day.
- Tubes shaken or moved during photos, mixing the layers and adding oxygen.
- Any mislabelled tube, spill, or missed recording.

3. How the errors may have affected the results

- **Low pH acting on the indicator:** could keep the dye blue even if some bacteria were growing. This would make vinegar look more effective than it was.
- **Curdling:** dye and oxygen may have been trapped away from bacteria in the curds. The dark blue liquid may have stayed blue because of this and not because growth stopped. This also overstates the effect.
- **Dilution of the milk:** tubes with more vinegar had less milk and less food per mL, so slower growth could be partly dilution and not only pH.
- **Foil vs rubber stoppers:** foil on the controls let in more oxygen, which would delay the control from turning white. This works against our result, so the real difference may be larger.
- **Daily photos and judging by eye:** could shift a result by about a day. The gap between the control and the vinegar tubes was larger than that, so the main conclusion should still hold.
- **One trial per condition:** one odd tube could distort a result, so the conclusion is less certain.

Overall, the errors mostly push toward making vinegar look better at stopping bacteria, so we should treat our result as supportive, not proven.

4. Value and relevance of the findings

- It shows that acidity can slow bacterial growth, which is why vinegar is used to pickle and preserve foods.
- It shows that methylene blue can work as a simple, cheap indicator of bacterial activity.
- It relates to food safety and shelf life, since acidic foods spoil more slowly.
- The curdling is a finding in itself: acid denatures the casein proteins in milk, the same process used in making cheese and yogurt.

5. Link to background information

- Milk is slightly acidic (pH about 6.5) and vinegar is about 2.5, so more vinegar lowers the pH further.
- Low pH damages bacterial proteins and cell membranes, which can slow growth or kill cells. Our results matched this: milk with vinegar did not turn white.
- Bacteria use up oxygen as they grow, which is why the indicator turns white in the milk-only control.

6. Next steps

- **Improve the design:** keep the milk-to-liquid ratio the same by adding water to reach the same total volume in every tube, so only acidity changes. Measure pH with pH strips or a probe, use the same cover on every tube, and run 3 trials per condition to average the results.
- **Extend the experiment:** test a vinegar-plus-indicator tube with no milk to check whether vinegar changes the dye on its own. Also try different acids at the same pH (for example vinegar vs lemon juice), and use a lower amount of vinegar (1 to 3 mL) so the tubes might not all stay blue and the amounts could be ranked.
- Add less vinegar to see the growth of bacteria in smaller volumes of vinegar like 1 or 2 ml

Quest study checklist

The Quest is a short written assessment (under 20 marks). Be able to state each of these without looking:

Question	Answer
Independent variable	Volume of vinegar added (5, 10, 15 mL)
Dependent variable	Days for the indicator to turn white
Control tubes	Milk only (no vinegar) and water only (no milk)
Controlled variables	Milk type and volume (15 mL), time in water bath (10 min), vinegar concentration (5%), indicator drops (10), dark storage, same tube cover

Question	Answer
Hypothesis	If vinegar is added to milk, bacterial growth slows because vinegar lowers the pH and harms cells
Result	Milk-only turned white in about 3 days; all vinegar tubes stayed blue and curdled
Supported?	Yes, vinegar slowed growth (but 5, 10 and 15 mL could not be ranked)
What the indicator means	White = oxygen used up by bacteria; blue = oxygen still present
Biggest errors	Vinegar may change the dye directly; curdling; milk dilution; foil vs stoppers; one trial; photo timing
Next step	Equal total volumes, pH measurement, 3 trials, a vinegar-and-dye tube with no milk

Remember the three error types: **systemic** is built into the method or equipment, **random** varies unpredictably, and **blunder** is a correctable mistake.